European Polymer Journal 223 (2025) 113678

Contents lists available at ScienceDirect

European Polymer Journal

journal homepage: www.elsevier.com/locate/europolj

Research Insights

Calcium-magnesium systems as lithium-free initiators for the anionic polymerization of butadiene

Thomas La Porta [a,b], Aurélie Michelet [a,b], Renaud Perrin [b], Mélanie Ray [b], Benoit Gadenne [b], Charlotte Dire [b], Stéphane Carlotti [a,*]

[a] Univ. Bordeaux, CNRS, Bordeaux INP, LCPO, UMR 5629, Pessac F-33600, France
[b] Manufacture Française Des Pneumatiques Michelin Centre de Technologie / Ladoux, 23 Place Des Carmes Déchaux, 63040 Clermont-Ferrand Cedex, France

ARTICLE INFO

Keywords:
Anionic polymerization
Butadiene
Calcium
Magnesium

ABSTRACT

Lithium-based initiators are commonly used for the polymerization of aromatic vinylic monomers and conjugated dienes. The corresponding polymers and copolymers are of huge importance in particular in the field of tire industry. Nevertheless, lithium is nowadays increasingly used for energy-related applications and alternatives has to be proposed in the field of anionic polymerization. A lithium-free anionic polymerization of butadiene using a new bimetallic initiator based on two alkaline earth metals, calcium and magnesium, is proposed and discussed. From two inactive initiators for butadiene polymerization, i.e $[(THF)_2Ca\{N(SiMe_3)_2\}_2]$ and [(n,s) Bu_2Mg], an active bimetallic initiator complex is prepared by mixing the two in apolar medium at 40 or 60 °C. Polybutadienes are synthesized with controlled molar masses, a low dispersity, and a living character. Interestingly, unusual microstructure of 75 % of 1,4-trans units with low vinyl content (<10 %) could be reached.

1. Introduction

In 1956, Szwarc [1,2], relying on his work and other researchers like Ziegler [3–5], Abkkin and Medvedev [6], demonstrated and named the living character of anionic polymerization using high vacuum techniques and permit to fully appreciate the potentiality of those systems. The rise of anionic polymerization and the main discoveries began from this date. Alkyllithium derivatives are mostly used as initiators for the anionic polymerization of vinyl aromatic and conjugated dienic monomers [7–9]. Due to the growth of lithium batteries and the lithium demand worldwide, there is an interest in finding alternatives to this alkali metal to anticipate its scarcity, its increasing cost and the increasing complexity of the supply chain. Looking at the use of other alkali and alkaline earth metal derivatives as anionic initiators, ring-opening polymerizations of cyclic esters are predominantly reported [10]. In particular calcium derivatives can be employed [11–15]. However, they are much less used for conjugated dienes and styrene polymerization. Magnesium derivatives are commonly used and usually combined with lanthanides, like neodymium. Researchers such as Visseaux, Zinck and coll. developed dialkylmagnesium:lanthanide systems for styrene [16–19] polymerization and conjugated dienes [20,21]. The dialkylmagnesium was used as a chain transfer agent to a lanthanide as a single site catalyst to conduct the coordinative chain transfer polymerization. The amounts of magnesium have an influence on molar masses, dispersities and catalytic activities. These systems were used in particular for the synthesis of stereospecific polydienes [22,23]. Focused on 1,4-trans dienes microstructure, Visseaux and coll. were pioneers for the synthesis of 1,4-trans polyisoprene using the combination of borohydride neodymium complexes with dialkylmagnesium in good yields [24,25] and more generally with polydienes like myrcene [26,27] or butadiene [28]. Looking at an anionic mechanism, only a few reports refer to the use of dialkylmagnesium as initiator/co-initiator for the polymerization of conjugated dienes. Dibutylmagnesium alone cannot initiate the butadiene or styrene polymerization, but, combined with an alkyllithium, initiation and polymerization occur [29–31]. It is observed, depending on [Mg]/[Li] ratios and the polarity of the medium, that the rate of the polymerization is decreased, as a proof of the formation of magnesiate complexes which reduce the reactivity of the active chain ends. Interestingly, in the case of secondary dialkylmagnesium, the molar masses of a polybutadiene obtained is lower than the expected one due to a co-initiation coming from the dialkylmagnesium. The bimetallic initiator does not affect the stereochemistry generally obtained with an alkyllithium. We have also shown recently that the association of calcium and lithium derivatives can

* Corresponding author.
E-mail address: carlotti@ensmac.fr (S. Carlotti).

https://doi.org/10.1016/j.eurpolymj.2024.113678
Received 8 October 2024; Received in revised form 17 December 2024; Accepted 18 December 2024
Available online 19 December 2024
0014-3057/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

promote the synthesis of high 1,4-trans polybutadienes with controlled molar masses, narrow dispersities, and living character [32,33].

The use of lithium-free initiator for anionic polymerization of styrene and conjugated dienes is even more challenging. A few reports mention possible solutions. As an example sodium-tributylmagnesiate complex is prepared from butylsodium and dibutylmagnesium [34]. Thanks to the dialkylmagnesium, the complex is soluble and stable enough to initiate the polymerization of isoprene. The polymerization appears to be not controlled with broad dispersities due to chain transfer and chain branching reactions. Heavier alkaline earth metals such as calcium-, strontium-, or barium-based complexes were proposed for the polymerization of styrene and dienes. The synthesis of [bis(phenyldimethylcarbinyl)barium] [35] and dibenzylbarium [36] are reported, and can be used as an anionic initiator for the polymerization of styrene. A barium-alkoxide with dialkylmagnesium is reported to produce polybutadiene with poor yields [37]. Adding 1,1-diphenylethylene (DPE) permits to generate an active barium complex allowing the preparation of polybutadienes with better yields. As an example, a polybutadiene with 72 % of 1,4-trans, 9 % of 1,2 and 19 % of 1,4-cis units with a yield of 100 % is obtained. Strontium and calcium chemistry have also been explored by Feil and Harder with the synthesis of [(*2-Me$_2$N-α*-Me$_3$Si-benzyl)Ae·(THF)$_2$] (Ae: Ca or Sr) [38,39]. The strontium analog shows a better reactivity and a lower dispersity than the calcium derivative. To enhance the reactivity and stability of calcium derivatives, a ligand exchange was applied to modify the steric hindrance. These new calcium complexes are active for the anionic polymerization of styrene and showed that the syndiotacticity is increased with the steric hindrance of the ligand [40]. Broad dispersities are generally obtained no matter the calcium derivatives used. Looking at the polymerization of butadiene initiated by calcium derivatives, Lindsell et al. [41] published Grignard-like initiators, i.e. Ph$_3$CCaCl(THF)$_2$ and Ph$_3$CCaBr(THF)$_4$, which were shown to initiate butadiene polymerization in THF at low temperature (−10 °C), yielding polybutadienes with molar masses around 10,000 g.mol^{-1}, and dispersities around 2.

It is worth mentioning that alkali metal derivatives, in particular alkylithium, are used to polymerize conjugated dienes, but also to synthesize the alkaline earth-based initiators. Herein, we report an innovative anionic alkali-free system, based on an unexpected synergy between calcium and magnesium derivatives for butadiene polymerization until full conversion. The control and livingness of the anionic polymerization of butadiene, and unusual microstructure of the resulting polybutadienes, is discussed.

2. Experimental section

2.1. Materials

Butadiene (99.5 % from Shell Chemicals), distilled under vacuum and stored under vacuum at room temperature in graduated glass tubes until use. Hexamethyldisilazane (HMDS, 99 %), calcium granular (99 %) and n,s-dibutylmagnesium ((n,s)Bu$_2$Mg) solution in hexane (0.7 mol. L^{-1}) were purchased from Sigma Aldrich and used without further purification. Cyclohexane (99 %, J.T. Baker) was purified by polystyryllithium seeds, distilled under vacuum, and then stored in graduated glass tubes under vacuum until use. Deuterated benzene (Sigma Aldrich) was dried over sodium/benzophenone and freshly distilled directly in a Young tube for NMR purpose·THF (Sigma Aldrich, 99.9 %) was dried over sodium/benzophenone, distilled under vacuum and stored under vacuum at room temperature in graduated glass tubes until use. Calcium bis[bis(trimethylsilyl)amide] ([(THF)$_2$Ca{N(SiMe$_3$)$_2$}$_2$]) was synthesized according to literature [32].

([(THF)$_2$Ca{N(SiMe$_3$)$_2$}$_2$]): ^{1}H NMR (400 MHz, C$_6$D$_6$, 25 °C): δ 3.54 ppm (t, 8H, THF), δ 1.21 ppm (t, 8H, THF), δ 0.37 ppm (s, 36H, N(SiMe$_3$)$_2$). ^{13}C NMR (400 MHz, C$_6$D$_6$, 25 °C): δ 69.7, 24.9 ppm (THF), δ 5.87 ppm (N(SiMe$_3$)$_2$).

2.2. Procedures

All polymerizations were performed at 40 °C or 60 °C under argon in a high-pressure glass reactor equipped with a magnetic stirrer and fitted with PTFE stopcocks previously flamed under vacuum. The calcium complex ([(THF)$_2$Ca{N(SiMe$_3$)$_2$}$_2$]) was weighed under glovebox, then added in a schlenk previously flamed under vacuum. Cyclohexane can be introduced under vacuum through connected glass tubes to obtain a solution at the desired concentration. A desired amount of calcium was injected in a schlenk previously flamed under vacuum followed by (n,s) Bu$_2$Mg in hexane via a syringe under argon flow. Cyclohexane was added under vacuum through connected glass tubes to obtain the desired concentration. And finally, butadiene was introduced under vacuum through connected glass tubes to start the polymerization. After the desired reaction time, degassed methanol was added to stop the reaction, and the possible remaining butadiene and cyclohexane were stripped off under vacuum. Conversions were determined using ^{1}H NMR (integration of butadiene vs. polybutadiene) with an aliquot taken during the run of polymerization (Fig. S1 for details).

Polybutadiene: ^{1}H NMR (400 MHz, CDCl$_3$, 25 °C): δ 5.57 ppm (m, 1H, vinyl proton from 1,2 unit), δ 5.43 ppm (s, 2H, vinyl protons from 1,4-trans unit), δ 5.39 ppm (s, 2H, vinyl protons from 1,4-cis unit), δ 4.96 ppm (m, 2H, vinyl protons from 1,2 unit), δ 2.05–208 ppm (2 s, 8H, protons from 1,4 unit), δ 0.8 ppm (m, 3H, protons from 1,2 unit).

As an example, for [Ca]/[Mg] = 0.75 (molar ratio), 53 mg of [(THF)$_2$Ca{N(SiMe$_3$)$_2$}$_2$] (105 μmol) was introduced into the glass reactor followed by 9 mL of cyclohexane. 200 μL of (n,s)Bu$_2$Mg (140 μmol) solution in hexane (0.7 M) was then added via a syringe under argon flow followed by 2 mL of liquid butadiene (22.8 mmol) to run the polymerization, then heated at 60 °C (run 11 Table 1).

The analysis of the calcium:magnesium complex at [Ca]/[Mg] = 0.75 ratio in cyclohexane solution (5–10 mL) was carried out by ^{1}H and ^{13}C NMR in deuterated benzene. An aliquot of the complex solution was first dried in a Young tube under vacuum to remove all volatiles before adding the deuterated solvent under vacuum (spectra are given in ESI Figs. S5 and S6).

2.3. Analysis

Polymer molar masses were determined by Size Exclusion Chromatography (SEC) using tetrahydrofuran (THF) as the eluent. Measurements in THF were performed on an Ultimate 3000 system equipped with differential refractive index detector dRI from Wyatt Technology. Polymers were separated on three G2000, G3000 and G4000 TOSOH HXL gel columns (300 × 7.8 mm) (exclusion limits from 1000 Da to 400 000 Da) at a flowrate of 1 mL/min. Columns temperature was held at 40 °C. Polybutadiene molar masses were determined using polystyrene standards and application of a corrective factor equal to 0.6 for low molar masses (<30,000 g.mol^{-1}) or 0.52 for high molar masses determined elsewhere.

Liquid-state ^{1}H NMR (400.2 MHz) and ^{13}C NMR (Inversed gated decoupling, 100.6 MHz, D1 = 10s, 4,096 scans) spectra were recorded at 298 K on a Bruker Avance 400 spectrometer in CDCl$_3$ or in C$_6$D$_6$. A detailed description of the method used to determine polybutadiene microstructure is presented in ESI Fig. S2.

MALDI-MS spectra were performed by the CESAMO (Bordeaux, France) on a Autoflex maX TOF mass spectrometer (Bruker Daltonics, Bremen, Germany) equipped with a frequency tripled Nd:YAG laser emitting at 355 nm. Spectra were recorded in the positive-ion mode with an accelerating voltage of 19 kV. Samples were dissolved in THF at 10 mg/ml. The 9-nitroanthracen matrix solution was prepared by dissolving 10 mg in 1 ml of THF. A THF solution of cationization agent (silver trifluoroacetate, 10 mg/ml) was also prepared. The solutions were combined in a 10:1:1 vol ratio of matrix to sample to cationization agent. One to two microliters of the solution obtained were deposited onto the sample target.

Table 1
Butadiene polymerization initiated by (n,s)Bu_2 Mg in the presence of different amounts of [$(THF)_2$ $Ca(HMDS)_2$] in cyclohexane.

Run	[Ca]/[Mg]	[M]/[I]	T (°C)	Time	Conv.[a] (%)	$\overline{M}_{n,th}$[b] ($g.mol^{-1}$)	$\overline{M}_{n,exp}$[c] ($g.mol^{-1}$)	Đ	1,2 (%)	1,4-trans (%)	1,4-cis (%)
1	0.5	181	40	24 h	68	6,600	4,600	1.07	11	75	14
2			60	24 h	99	9,700	5,700	1.22	10	73	17
3	0.75	181	40	24 h	78	7,600	5,200	1.05	10	75	15
4			60	24 h	100	9,800	7,000	1.10	10	74	16
5	1	181	40	24 h	80	7,800	6,900	1.08	11	75	14
6			60	24 h	98	9,600	7,300	1.21	10	73	17
7	1.25	181	40	24 h	80	7,800	7,300	1.13	11	75	14
8			60	24 h	99	9,700	8,300	1.15	10	73	17
9	1.5	181	40	24 h	84	8,200	8,000	1.15	10	74	16
10			60	24 h	99	9,700	9,200	1.21	10	73	17
11	0.75[d]	163	60	4d	100	8,800	8,100	1.11	10	74	16
12		481	60	4d	100	26,000	23,300	1.18	8	75	17
13		1945	60	6d	97	105,000	90,000	1.44	7	68	25

[a] Determined by ^{1}H NMR regarding butadiene vs. polybutadiene.
[b] Calculated with one polymer chain per Mg at the given conversion.
[c] Number-average molar mass estimated by SEC based on polystyrene standards applying a corrective factor (0.6 for low molar masses and 0.52 for high molar masses [last experiment]).
[d] Polymerization run in closed reactor without withdrawal until quenched.

Inductively Coupled Plasma analysis was performed by the Laboratoire de Chimie de Coordination (Toulouse, France) on ICP-OES ICAP 6300 Duo Thermo. Wavelength range: 166–847 nm; detector: high performance CID86 chip; RF source: 27.12 MHz solid state, 750–1500 W; plasma gas: fixed 12L.min^{-1}, argon; nebulizer gas: pressure control from 0 to 0.4 MPa.

3. Results and discussions

Based on the work of Krieck et al. [42], [$(THF)_2Ca\{N(SiMe_3)_2\}_2$] complex was first synthesized and characterized by ^{1}H NMR in C_6D_6 (Fig. S3). Inductively Coupled Plasma analysis showed the presence of a tiny number of impurities, mainly calcium dibromide (Table S1). This complex was then used without further purification as this impurity is not expected to terminate the butadienyl active centers. This was confirmed by preliminary experiments. Butadiene polymerization was then studied using [$(THF)_2Ca\{N(SiMe_3)_2\}_2$] with (n,s)$Bu_2$Mg as initiator in different proportions. Results are given in Table 1.

As shown, butadiene polymerized using different ratios of [Ca]/[Mg], ranging from 0.5 to 1.5, with narrow or quite narrow chain distributions depending on this ratio. The lowest dispersity was obtained for [Ca]/[Mg] = 0.75 at 40 or 60 °C, and it tends generally to increase when temperature increases from 40 to 60 °C. One can note that molar masses are for most samples lower than the expected ones due to repetitive opening of the reactor for aliquot withdrawals used for kinetics following and causing a loss of butadiene. Polymerizations carried out in closed reactors at 60 °C confirms that molar masses can be closer to the theoretical values as shown in run 11 and 12. They are still lower, possibly due to an initiation by the second alkyl group of the Mg derivative giving more than 1 chain per "ate" complex. Looking at the dispersities, a slow initiation of the complex, as compared to the propagation step, might be also considered when increasing the reaction temperature or when a high molar mass is targeted. Those experiments also confirmed that full conversion can be reached, and high molar mass as well with run 13.

As known in anionic polymerization, the 1,4-trans microstructure is reduced when the concentration of active species is reduced [43], due to different rates of chain ends conformations of active species (Fig. S4) with $k_{cis}/k_{trans} = 2$ for butadiene [44]. Indeed, in these conditions, the propagation rate is favored as compared to the isomerization rate, yielding to an increasing amount of 1,4-cis units. The decrease of the [Ca]/[Mg] initiator concentration by 10 as compared to the other experiments confirms that observation (Table 1, run 13). High resolution MALDI-TOF MS analysis showed that a C_4H_9 group initiated the polymerization and a H ended it. As shown in Fig. 1a), molar masses obtained for $\overline{DP}_n = 14$ agree with theoretical ones simulated by an isotopic pattern (monoisotopic mass equal to 921.64 g.mol^{-1} and an average of 923.65 g.mol^{-1}). A butyl group provided by the dialkylmagnesium initiates the polymerization and the theoretical molar masses were calculated according to the amount of dialkylmagnesium with 1 chain per magnesium. Moreover, one population of polymer is observed and confirmed that no transfer reaction to the complexed THF occurs (Fig. 1b)).

To understand and represent the structure of the initiating complex, ^{1}H NMR was first carried out (Fig. S5). The spectrum of the Ca:Mg complex showed that most of the peaks present in the initial (n,s)Bu_2Mg are shifted meaning that the magnesium derivative has reacted with the calcium one to form a bimetallic complex. However, the exact structure is unclear. As shown in HSQC experiment (Fig. S6), the amide ligand from calcium gives several singlets between 0 to 0.6 ppm which agrees that calcium is involved in bimetallic complexes. This can be interpreted by different calcium-magnesium stoichiometries and by different geometries between amide-calcium and amide-magnesium derivatives. Based on some literature on "ate" complexes[30,31], it is proposed that the initiator is formed via a 4-membered ring as described in Scheme 1.

Interestingly, the microstructure of the prepared polybutadiene from Ca:Mg systems is mainly 1,4 units with low 1,2 units (Table 1), similarly as a polybutadiene obtained with an alkyllithium in apolar condition [45]. As a difference, monometallic alkyllithium allows to reach up to 55 % of 1,4-trans units, when 73–75 % are obtained with this bimetallic system whatever the ratio between calcium and magnesium. Such values are in agreement with the microstructure observed with a Ca:Li system

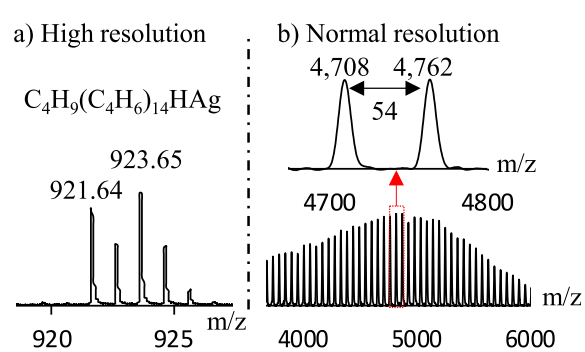
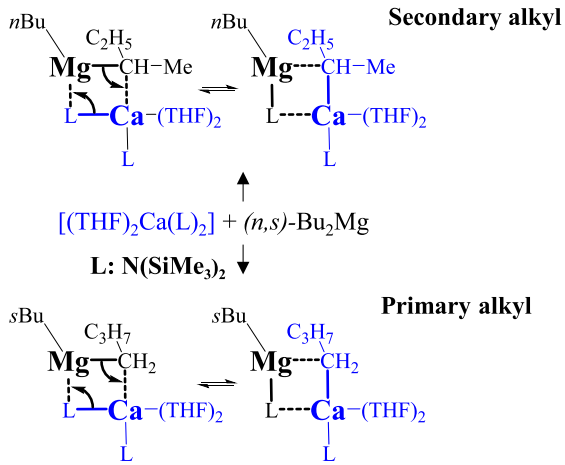


Fig. 1. MALDI-TOF MS of polybutadiene obtained with [$(THF)_2$ $Ca\{N(SiMe_3)_2\}_2$]:(n,s)Bu_2 Mg ([Ca]/[Mg] = 0.75, 60 °C, cyclohexane).



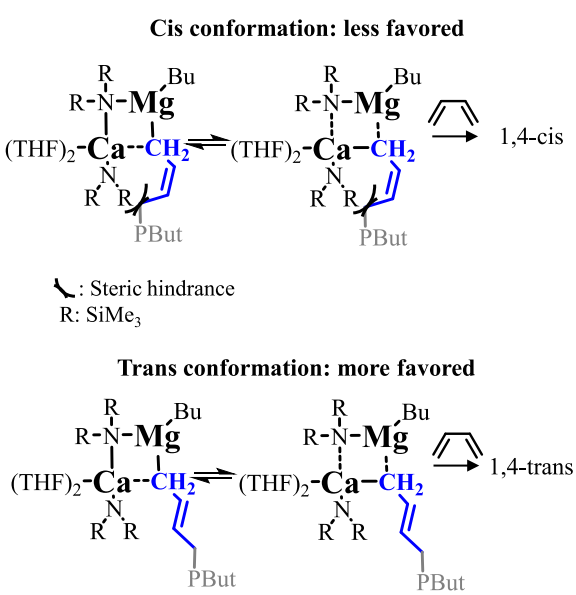
Scheme 1. Proposed mechanism for the formation of an active butyl-calcium bond within the Ca:Mg complex.

previously published [32,33]. C-Ca bonds, in situ generated, act as propagative species, yielding to high 1,4-trans units (Scheme 2). The equilibrium between syn- and anti-conformation of an active chain ends is shifted to *syn*-conformation due to the steric hindrance induced by bis-trimethylsilyl-amide ligands.

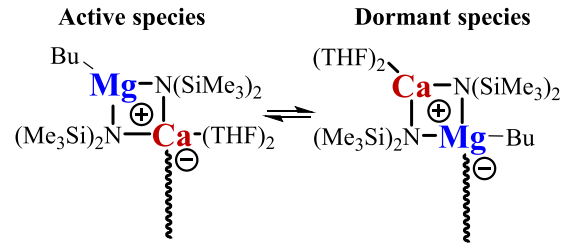
It is worth adding that an equilibrium between active C-Ca bonds and dormant C-Mg ones can also exist (Scheme 3), which explain the limited possibility to modulate the 1,4-cis / 1,4-trans ratio (Table 1). This can be related to the possibility to increase the 1,4-cis, or decrease the 1,4-trans units, when Ca:Li systems are used. Indeed, both C-Ca and C-Li bonds are active meaning that, when [Ca]/[Li] ratios decrease, C-Li bonds are predominant yielding to a decrease of the 1,4-trans content.

Isoprene was also polymerized with the Ca:Mg system to confirm the nature of the active species yielding mostly 1,4-trans units (Table S2). As compared to the 10–30 % (depending on [M]/[I] ratios) of 1,4-trans units obtained from n-BuLi initiation [46], polyisoprenes obtained from the Ca:Mg initiator are mainly composed of 1,4-trans units (55–58 %). These results agree with the proposed mechanism based on an active C-Ca bond.

Polymerization kinetics were also determined using $-d[\mathrm{But}]/dt = R_p = k_p[\mathrm{Ca:Mg}]_0[\mathrm{But}] = k_{app}[\mathrm{But}]$, with k_{app} the slope of the fitted curves



Scheme 2. Structures of polybutadiene propagating species yielding 1,4-trans units.



Scheme 3. Equilibrium between active C–Ca bond and non-active C–Mg bond.

given by the equation $-\ln(1-\mathrm{Conv.}) = k_{app}\,t$. The influence of [Ca]/[Mg] ratios between 0.5 and 1.5 on the propagation rate was first studied at 60 °C (Fig. 2a)). The sub-stoichiometry of one metal (i.e. [Ca]/[Mg] = 0.75 or 1.25) allows the propagation rate to increase slightly. This was not observed at 40 °C (Fig. S7). Now, comparing the kinetics, it can be noted that k^1_{app} is almost divided by 2 at 60 °C, and more than 10 at 40 °C (k^2_{app}) using [Ca]/[Mg] = 0.75 as compared to the *n*-BuLi reference k^0_{app} at 40 °C (Fig. 2b)). The new Ca:Mg active species are less reactive than alkyllithium derivatives used alone due to the delocalization of the anionic charge within the Ca:Mg complex. This supports the presented equilibrium between active and dormant species.

The evolution of molar masses with conversion and dispersities was plotted for [Ca]/[Mg] = 0.5, 0.75, 1, 1.25 and 1.5 at 60 °C. The absence of transfer and termination reactions is confirmed by the linear evolution for all systems. As illustrated in Fig. 3 (and Fig. S8 at 40 °C), the evolution of molar masses with conversion are quite linear no matter the [Ca]/[Mg] ratios, which agrees with the absence of side reactions. It is confirmed by MALDI-TOF (Fig. 1) and SEC (Fig. S9) with only 1 population of polymers observed. One can note that the dispersities increase up to 1.2 after 50 % of conversion, except for a small sub-stoichiometric amount of calcium, i.e. [Ca]/[Mg] = 0.75 (Fig. 3) and is explained by aggregation issues.

The livingness of Ca:Mg initiated polymerization of butadiene was investigated by a sequential addition of butadiene at 60 °C with the most efficient system, i.e. for a [Ca]/[Mg] ratio equal to 0.75. As shown in Fig. S10, when butadiene is added to the medium after full conversion of the first batch, not all chains could be re-initiated. However, if the second addition of butadiene is added during the polymerization process of the first batch, i.e. before reaching full conversion, almost all chains could be re-initiated as shown in Fig. 4. The same phenomenon was observed in our previous work using Ca:Li systems[32]. As described with these calcium-lithium systems, similar aggregation issues can explain this phenomenon due to dormant chain ends hypothesis.

The nature of alkyl group from magnesium derivatives was also investigated using (n,s)Bu_2Mg, (n,n)Bu_2Mg and n-Bu,n-OctMg (Table 2). As expected, all systems reached full conversion with a similar reaction time and with controlled molar masses. Looking at the dispersities of the polybutadienes obtained, (n,s)Bu_2Mg-based initiating complexes yield the lowest ones. The secondary alkyl of the magnesium derivative seems to allow an easier ligand exchange into to the Ca:Mg complex for the formation of C–Ca bond.

4. Conclusion

The combination of calcium and magnesium derivatives is an efficient route to create lithium-free initiating systems for the anionic polymerization of butadiene. The polybutadienes are obtained with controlled molar masses, narrow dispersities and low 1,2 units content (≤10 %). Interestingly, unusual high 1,4-trans microstructure (75 %) was reached whatever the [Ca]/[Mg] ratio used. Those results confirm the role of the calcium compound within the "ate" complex with the formation of an active C-Ca bond. Furthermore, it is important to highlight that no alkali metal was required as an alkyl donor, the magnesium derivative supplying the alkyl group.

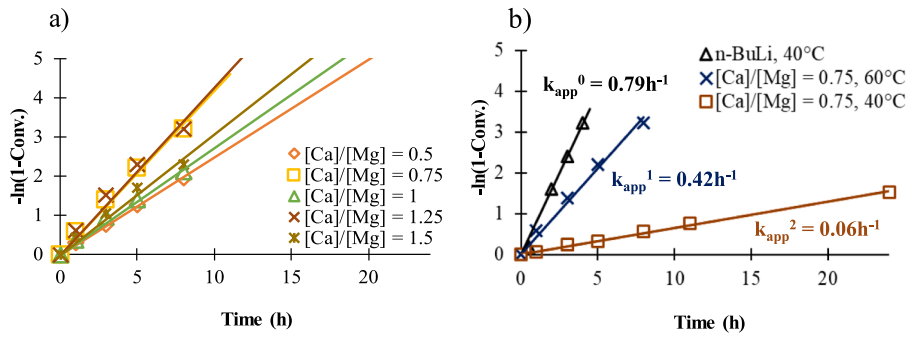


Fig. 2. (a) $\ln[M]/[M]_0$ vs time plots for the polymerization of butadiene initiated by bimetallic Ca:Mg systems at 60 °C, (b) Comparison of k_{app} of butadiene polymerization initiated by n-BuLi ($[\text{n-BuLi}]_0$ = 13.8 mmol.L^{-1}) vs. Ca:Mg systems ($[Mg]_0$ = 13.8 mmol.L^{-1}, [Ca]/[Mg] = 0.75) (cyclohexane, $[\text{But}]_0$ = 2.5 mol.L^{-1}, $\overline{M}_{n,th}$ = 9,800 g.mol^{-1}).

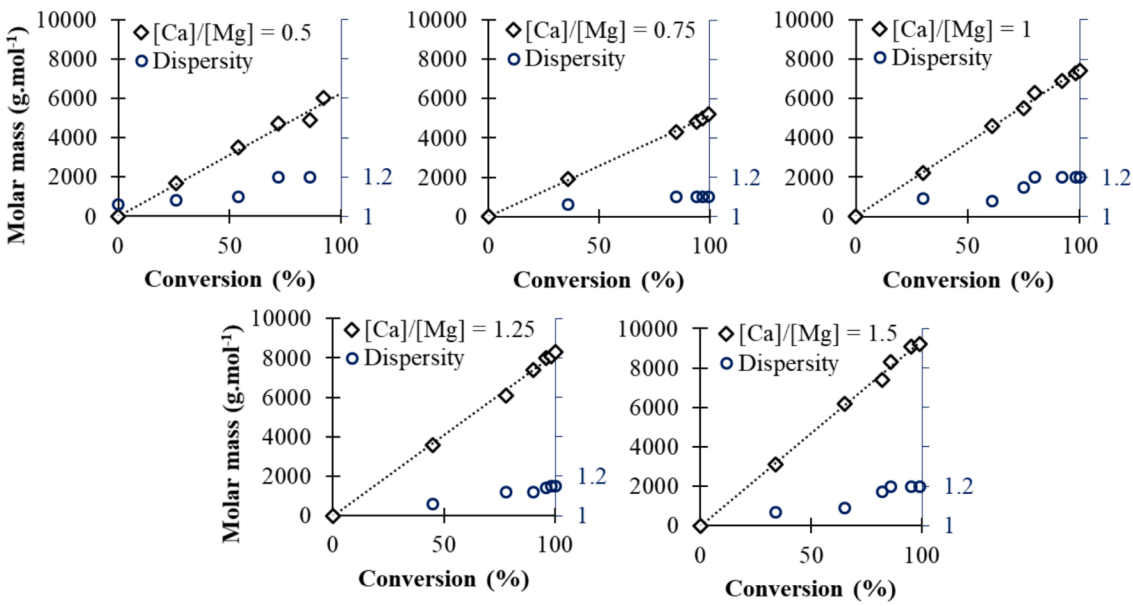


Fig. 3. Molar mass and dispersity vs. conversion plots with fitted curve for the polymerization of butadiene initiated by bimetallic Ca:Mg systems ($\overline{M}_{n,th}$ = 9,800 g.mol^{-1}, cyclohexane, 60 °C).

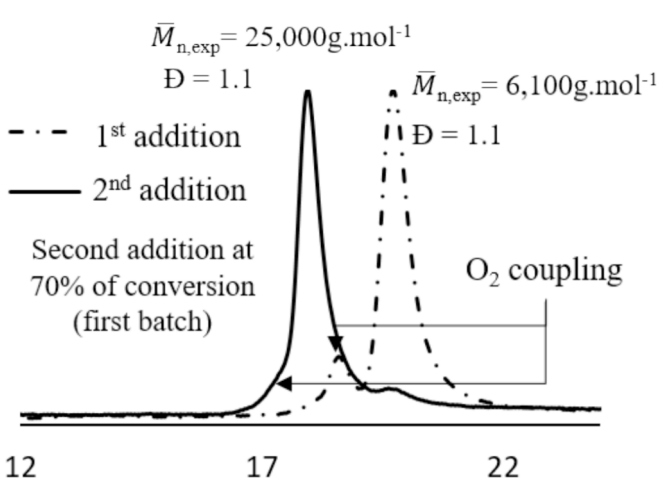


Fig. 4. SEC of a polybutadiene initiated by $[(THF)_2\ Ca\{N(SiMe_3)_2\}_2]$:(n,s)$Bu_2$Mg system ([Ca]/[Mg] = 0.75, cyclohexane, 60 °C). Butadiene was added two times, first addition with $\overline{M}_{n,th}$ = 9,000 g.mol^{-1} as targeted mass, then a second addition to reach $\overline{M}_{n,th}$ = 18,000 g.mol^{-1}.

The best [Ca]/[Mg] ratio appeared to be 0.75 as a good compromise can be obtained with full conversion, controlled molar masses by the amount of (n,s)Bu_2Mg introduced, low dispersity (D = 1.1), a living character of the butadiene polymerization and a relative fast polymerization at 60 °C. This calcium:magnesium system is still lower as compared to alkyllithium initiators in terms of polymerization rate of butadiene but, opens a new era in the field of lithium-free anionic initiators for conjugated dienes and copolymers.

CRediT authorship contribution statement

Thomas La Porta: Writing – review & editing, Writing – original draft, Methodology, Visualization, Validation, Investigation, Formal analysis, Data curation. **Aurélie Michelet:** Data curation, Investigation, Methodology, Validation, Visualization, Writing – review & editing. **Renaud Perrin:** Validation, Supervision, Conceptualization. **Mélanie Ray:** Validation, Supervision, Conceptualization. **Benoit Gadenne:** Validation, Supervision, Methodology, Funding acquisition, Conceptualization. **Charlotte Dire:** Validation, Supervision, Resources, Project administration, Methodology, Conceptualization. **Stéphane Carlotti:** Writing – review & editing, Validation, Supervision, Resources, Project

Table 2
Butadiene polymerization initiated by different dialkylmagnesium derivatives in the presence of various amounts of $[(THF)_2Ca\{N(SiMe_3)_2\}_2]$ in cyclohexane at 60 °C during 24 h.

[Ca]/[Mg]	Alkylmagnesium	Conv.[a] (%)	$\overline{M}_{n,th}$[b] (g. mol^{-1})	$\overline{M}_{n,exp}$[c] (g. mol^{-1})	Đ
0.5	(n,s)Bu_2Mg	100	11,000	8500	1.1
	(n,n)Bu_2Mg	100	12,300	11,400	1.4
	n-Bu-n-OctMg	100	12,300	13,200	1.4
0.75	(n,s)Bu_2Mg	100	11,000	9900	1.1
	(n,n)Bu_2Mg	100	12,300	11,000	1.4
	n-Bu-n-OctMg	100	12,300	11,000	1.4
1	(n,s)Bu_2Mg	100	11,000	9,900	1.1
	(n,n)Bu_2Mg	100	12,300	11,400	1.3
	n-Bu-n-OctMg	100	12,300	11,400	1.4
1.25	(n,s)Bu_2Mg	100	11,000	9600	1.1
	(n,n)Bu_2Mg	100	12,300	11,200	1.4
	n-Bu-n-OctMg	100	12,300	12,600	1.2
1.5	(n,s)Bu_2Mg	100	11,000	9700	1.2
	(n,n)Bu_2Mg	100	12,300	11,400	1.3
	n-Bu-n-OctMg	100	12,300	13,800	1.3

[a] Determined by ^{1}H NMR regarding butadiene vs. polybutadiene.
[b] Calculated with one polymer chain per Mg.
[c] Number-average molar mass estimated by SEC on the basis of polystyrene standards applying x0.6 corrective factor.

administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.113678.

Data availability

No data was used for the research described in the article.

References

[1] M. Szwarc, Living, PolymersNature 178 (4543) (1956) 1168–1169, https://doi.org/10.1038/1781168a0.
[2] M. Szwarc, M. Levy, R. Milkovich, Polymerization initiated by electron transfer to monomer. A new method of formation of block polymers, J. Am. Chem. Soc. 78 (11) (1956) 2656–2657, https://doi.org/10.1021/ja01592a101.
[3] K. Ziegler, K. Bähr, D. Über, Vermutlichen mechanismus der polymerisationen durch alkalimetalle (vorläufige mitteilung), Berichte Der Deutschen Chem. Gesellschaft (A and B Series) 61 (2) (1928) 253–263, https://doi.org/10.1002/cber.19280610203.
[4] K. Ziegler, F. Crössmann, H. Kleiner, O. Schäfer, Untersuchungen über alkali-organische verbindungen. I. reaktionen zwischen ungesättigten kohlenwasserstoffen und alkalimetall-alkylen, Justus Liebigs Ann. Chem. 473 (1) (1929) 1–35, https://doi.org/10.1002/jlac.19294730102.
[5] K. Ziegler, Die Bedeutung Der Alkalimetallorganischen Verbindungen Für Die Synthese, Angew. Chem. 49 (28) (1936) 455–460, https://doi.org/10.1002/ange.19360492807.
[6] A. Abkin, S. Medvedev, On the kinetics of the polymerisation of 1.3 butadiene in the presence of metallic sodium, Trans. Faraday Soc. 32 (1936) 286, https://doi.org/10.1039/tf9363200286.
[7] H.S. Makowski, M. Lynn, B.A.N. Butyllithium, Polymerization of butadiene. II. Effects of solvent and temperature on association of polybutatadienyl lithium. J. Macrom. Sci.: Part A - C Chem. 2 (4) (1968) 665–682, https://doi.org/10.1080/10601326808051433.
[8] A. Yamagishi, M. Szwarc, Kinetics of styrene addition in benzene solution to living lithium polymers terminated by 1,1-diphenylethylene units. the effect of mixed dimerization of monomeric polymers, Macromolecules 11 (3) (1978) 504–506, https://doi.org/10.1021/ma60063a016.
[9] H.L. Hsieh, R.P. Quirk, Anionic polymerization: Principles and practical applications, Plastics engineering; Marcel Dekker, New York, 1996.
[10] S. Carlotti, F. Peruch (Eds.), Cyclic Monomers: Epoxides, Lactide, Lactones, Lactams, Cyclic Silicon-Containing Monomers, Cyclic Carbonates, and Others. *Anionic Polymerization*; Hadjichristidis, N., Hirao, A., Springer Japan: Tokyo, 2015; pp 191–305. https://doi.org/10.1007/978-4-431-54186-8_5.
[11] Z. Zhong, P.J. Dijkstra, C. Birg, M. Westerhausen, J. Feijen, A Novel and versatile calcium-based initiator system for the ring-opening polymerization of cyclic esters, Macromolecules 34 (12) (2001) 3863–3868, https://doi.org/10.1021/ma0019510.
[12] M. Westerhausen, S. Schneiderbauer, A.N. Kneifel, Y. Söltl, P. Mayer, H. Nöth, Z. Zhong, P.J. Dijkstra, J. Feijen, Organocalcium compounds with catalytic activity for the ring-opening polymerization of lactones, Eur. J. Inorg. Chem. 2003 (18) (2003) 3432–3439, https://doi.org/10.1002/ejic.200300286.
[13] L. Piao, M. Deng, X. Chen, L. Jiang, X. Jing, Ring-opening polymerization of ε-caprolactone and l-lactide using organic amino calcium catalyst, Polymer 44 (8) (2003) 2331–2336, https://doi.org/10.1016/S0032-3861(03)00118-6.
[14] M.H. Chisholm, J.C. Gallucci, K. Phomphrai, Well-defined calcium initiators for lactide polymerization, Inorg. Chem. 43 (21) (2004) 6717–6725, https://doi.org/10.1021/ic0490730.
[15] R.A. Collins, J. Unruangsri, P. Mountford, Synthesis and rac-lactide ring-opening polymerisation studies of new alkaline earth tetrahydroborate complexes, Dalton Trans. 42 (3) (2013) 759–769, https://doi.org/10.1039/C2DT32151C.
[16] P. Zinck, M. Visseaux, A. Mortreux, Borohydrido rare earth complexes as precatalysts for the polymerisation of styrene, Z. Anorg. Allg. Chem. 632 (12–13) (2006) 1943–1944, https://doi.org/10.1002/zaac.200600143.
[17] P. Zinck, A. Valente, F. Bonnet, A. Violante, A. Mortreux, M. Visseaux, S. Ilinca, R. Duchateau, P. Roussel, Reversible coordinative chain transfer polymerization of styrene by rare earth borohydrides, chlorides/dialkylmagnesium systems: Coordinative chain transfer polymerization, J. Polym. Sci. A Polym. Chem. 48 (4) (2010) 802–814, https://doi.org/10.1002/pola.23828.
[18] A. Valente, P. Zinck, A. Mortreux, M. Visseaux, Borohydrido rare earth based coordinative chain transfer copolymerization: A convenient tool for tuning the microstructure of isoprene/styrene copolymers: Coordinative chain transfer copolymerization, J. Polym. Sci. A Polym. Chem. 49 (7) (2011) 1615–1620, https://doi.org/10.1002/pola.24585.
[19] A. Valente, A. Mortreux, M. Visseaux, P. Zinck, Coordinative chain transfer polymerization, Chem. Rev. 113 (5) (2013) 3836–3857, https://doi.org/10.1021/cr300289z.
[20] A. Valente, P. Zinck, A. Mortreux, M. Visseaux, Catalytic chain transfer (Co-) polymerization: Unprecedented polyisoprene CCG and a new concept to tune the composition of a statistical copolymer, Macromol. Rapid Commun. 30 (7) (2009) 528–531, https://doi.org/10.1002/marc.200800734.
[21] A. Valente, P. Zinck, M.J. Vitorino, A. Mortreux, M. Visseaux, Rare earths/main group metal alkyls catalytic systems for the 1,4-trans stereoselective coordinative chain transfer polymerization of isoprene, J. Polym. Sci. A Polym. Chem. 48 (21) (2010) 4640–4647, https://doi.org/10.1002/pola.24225.
[22] J. Gromada, L. Le Pichon, A. Mortreux, F. Leising, J.-F. Carpentier, Neodymium Alk (Aryl)oxides–dialkylmagnesium systems for butadiene polymerization and copolymerization with styrene and glycidyl methacrylate, J. Organomet. Chem. 683 (1) (2003) 44–55, https://doi.org/10.1016/S0022-328X(03)00380-2.
[23] P. Zinck, M. Terrier, A. Mortreux, A. Valente, M. Visseaux, A new family of styrene/diene rubbers, Macro Chem. Phys. 208 (9) (2007) 973–978, https://doi.org/10.1002/macp.200700055.
[24] F. Bonnet, M. Visseaux, A. Pereira, F. Bouyer, D. Barbier-Baudry, Stereospecific polymerization of isoprene with $Nd(BH_4)_3(THF)_3/MgBu_2$ as catalyst, Macromol. Rapid Commun. 25 (8) (2004) 873–877, https://doi.org/10.1002/marc.200300289.
[25] M. Terrier, M. Visseaux, T. Chenal, A. Mortreux, Controlled trans -stereospecific polymerization of isoprene with lanthanide(III) borohydride/dialkylmagnesium systems: The improvement of the activity and selectivity, kinetic studies, and mechanistic aspects, J. Polym. Sci. A Polym. Chem. 45 (12) (2007) 2400–2409, https://doi.org/10.1002/pola.22002.
[26] S. Loughmari, A. Hafid, A. Bouazza, A. El Bouadili, P. Zinck, M. Visseaux, Highly stereoselective coordination polymerization of B-myrcene from a lanthanide-based catalyst: Access to bio-sourced elastomers, J. Polym. Sci. A Polym. Chem. 50 (14) (2012) 2898–2905, https://doi.org/10.1002/pola.26069.
[27] S. Loughmari, A. Hafid, A. Bouazza, A. El Bouadili, P. Zinck, M.C. Visseaux, Highly stereoselective coordination polymerization of B-myrcene from a lanthanide-based catalyst: Access to bio-sourced elastomers. J. Polym. Sci. Part A Polym. Chem. 52 (11) (2014) 1642, https://doi.org/10.1002/pola.27157.
[28] A. Ventura, T. Chenal, M. Bria, F. Bonnet, P. Zinck, Y. Ngono-Ravache, E. Balanzat, M. Visseaux, Trans-stereospecific polymerization of butadiene and random copolymerization with styrene using borohydrido neodymium/magnesium dialkyl catalysts, Eur. Polym. J. 49 (12) (2013) 4130–4140, https://doi.org/10.1016/j.eurpolymj.2013.09.019.
[29] H.L. Hsieh, I.W. Wang, Effects of dibutylmagnesium on alkyllithium-initiated polymerizations, Macromolecules 19 (2) (1986) 299–304, https://doi.org/10.1021/ma00156a009.
[30] S. Ménoret, S. Carlotti, M. Fontanille, A. Deffieux, P. Desbois, C. Schade, W. Schrepp, V. Warzelhan, Retarded Anionic Polymerization, 5. Influence of the Structure of Dialkylmagnesium Additives on the Reactivity of Polystyryllithium Species. *Macromolecular Chemistry and Physics* **2001**, *202* (16), 3219–3227. https://doi.org/10.1002/1521-3935(20011101)202:16<3219::AID-MACP3219>3.0.CO;2-M.
[31] S. Carlotti, P. Desbois, V. Warzelhan, A. Deffieux, Retarded anionic polymerization (RAP) of styrene and dienes, Polymer 50 (14) (2009) 3057–3067, https://doi.org/10.1016/j.polymer.2009.05.015.

[32] T. La Porta, R. Perrin, M. Ray, B. Gadenne, C. Dire, S. Carlotti, Calcium–lithium systems as innovative bimetallic initiators for the anionic polymerization of butadiene: Toward control and high 1,4-trans microstructure, Macromolecules 57 (5) (2024) 2376–2384, https://doi.org/10.1021/acs.macromol.3c02160.

[33] K. Roos, S. Carloti, A. FORENS, B. Gadenne, Catalytic System for the Stereospecific Polymerization of Dienes and Use Thereof in a Process for Synthesizing Diene Polymers. US20230272128A1, August 31, 2023. https://patents.google.com/patent/US20230272128A1/en?q=(carloti)&assignee=michelin&oq=michelin+carloti (accessed 2024-06-04).

[34] M. Liu, C. Kamienski, M. Morton, L.J. Fetters, Anionic polymerization characteristics of the 1:1 complex of n-butylsodium and n-,s-dibutylmagnesium (sodium tributylmagnesiate), J. Macrom. Sci.: Part A – Chem. 23 (12) (1986) 1387–1396, https://doi.org/10.1080/00222338608081131.

[35] L. Chai Tang, C. Mathis, B.D. Francois, A novel initiator in anionic polymerization. Its physicochemical properties in tetrahydrofuran and tetrahydropyran, J. Organomet. Chem. 243 (4) (1983) 359–371, https://doi.org/10.1016/S0022-328X(00)98569-3.

[36] B. De Groff, M. Van Beylen, M. Szwarc, Anionic polymerization of the barium salt of one-ended living polystyrene I. conductance and kinetic studies, Macromolecules 8 (4) (1975) 396–400, https://doi.org/10.1021/ma60046a005.

[37] Z.M. Baidakova, E.V. Kristal'nyi, A.A. Arest-Yakubovich, Butadiene polymerization in the presence of initiators of the type barium alcoholates-organomagnesium compound, Polymer Science U.S.S.R. 18 (9) (1976) 2325–2330, https://doi.org/10.1016/0032-3950(76)90108-8.

[38] S. Harder, F. Feil, A. Weeber, Structure of a benzylcalcium diastereomer: An initiator for the anionic polymerization of styrene, Organometallics 20 (6) (2001) 1044–1046, https://doi.org/10.1021/om000945p.

[39] F. Feil, S. Harder, Benzyl complexes of the heavier alkaline-earth metals: The first crystal structure of a dibenzylstrontium complex, Organometallics 20 (22) (2001) 4616–4622, https://doi.org/10.1021/om010444j.

[40] F. Feil, C. Müller, S. Harder, α-Methyl-benzylcalcium complexes: Syntheses, structures and reactivity, J. Organomet. Chem. 683 (1) (2003) 56–63, https://doi.org/10.1016/S0022-328X(03)00405-4.

[41] W.E. Lindsell, F.C. Robertson, I. Soutar, Polymerization by alkaline earth metal compounds—II polymerization of 2- and 4-vinylpyridine, styrene and butadiene by triphenylmethyl derivatives of calciumStrontium and Barium, European Polymer J. 19 (2) (1983) 115–120, https://doi.org/10.1016/0014-3057(83)90024-1.

[42] S. Krieck, P. Schüler, J.M. Peschel, M. Westerhausen, Straightforward one-pot syntheses of silylamides of magnesium and calcium via an in situ grignard metalation method, Synthesis 51 (05) (2019) 1115–1122, https://doi.org/10.1055/s-0037-1610407.

[43] *Initiation of Polymerization*; Bailey, F. E., Vandenberg, E. J., Blumstein, A., Bowden, M. J., Arthur, J. C., Lal, J., Ottenbrite, R. M., Eds.; ACS Symposium Series; AMERICAN CHEMICAL SOCIETY: WASHINGTONG D.C., 1983; Vol. 212, Chapter 21, 283-289. https://doi.org/10.1021/bk-1983-0212.

[44] W.H. Glaze, J.E. Hanicak, M.L. Moore, J. Chaudhuri, 3-Neopentylallyllithium I. The 1,4-addition product of tert-butyllithium and 1,3-butadiene, J. Organomet. Chem. 44 (1) (1972) 39–48, https://doi.org/10.1016/0022-328X(72)80040-8.

[45] A. Forens, K. Roos, C. Dire, B. Gadenne, S. Carlotti, Accessible microstructures of polybutadiene by anionic polymerization, Polymer 153 (2018) 103–122, https://doi.org/10.1016/j.polymer.2018.07.071.

[46] H.L. Hsieh, Microstructures of polydienes prepared from alkyllithium, J. Polym. Sci., Part A: Gen. Pap. 3 (1) (1965) 181–190, https://doi.org/10.1002/pol.1965.100030120.